

Linear Algebra Project - Theory Guide

Part A: Vector & Matrix Fundamentals

- **Vector:** A vector is an ordered collection of values used to represent magnitude and direction in data.
- **Matrix:** A matrix is a rectangular arrangement of numbers used to organize and process datasets.
- **Norm-1:** Norm-1 measures the total absolute magnitude of a vector.
- **Norm-2:** Norm-2 represents the overall length or magnitude of a vector.
- **Dot Product:** The dot product measures the similarity between two vectors.
- **Cross Product:** The cross product produces a vector perpendicular to two three-dimensional vectors.
- **Vector Projection:** Vector projection shows how much of one vector lies in the direction of another vector.

Part B: Matrix Operations

- **Matrix Addition:** Matrix addition combines corresponding elements of matrices with the same dimensions.
- **Matrix Multiplication:** Matrix multiplication transforms and combines information from different matrices.
- **Transpose:** Transpose swaps rows and columns of a matrix.
- **Inverse:** The inverse matrix reverses the effect of a matrix transformation when it exists.
- **Determinant:** The determinant indicates important properties such as invertibility of a matrix.

Part C: Linear Transformations & Geometry

- **Line:** A line represents data using one dimension or feature.
- **Plane:** A plane represents data using two dimensions or features.
- **Hyperplane:** A hyperplane extends the concept of a plane into higher dimensions.
- **Dimensionality:** Dimensionality refers to the number of features used to describe data.

Part D: Eigenvalues & Decomposition

- **Eigenvalues:** Eigenvalues indicate the amount of variance captured in specific directions.
- **Eigenvectors:** Eigenvectors define the directions of important patterns in the data.
- **LU Decomposition:** LU decomposition breaks a matrix into lower and upper triangular matrices.
- **SVD:** Singular Value Decomposition factorizes a matrix to simplify analysis and dimensionality reduction.

Part E: Dimensionality Reduction

- **PCA:** PCA reduces dimensions while preserving most of the important information.
- **LDA:** LDA reduces dimensions by maximizing separation between predefined classes.